

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Lubricating Oil Cooler Pressure Test Kit, Part Number 3823876

Additional Service Items

- Brush
- Shop air
- Cleaning tank
- Screwdriver
- Gasket scraper

Preparatory Steps

All Applications Except Marine

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

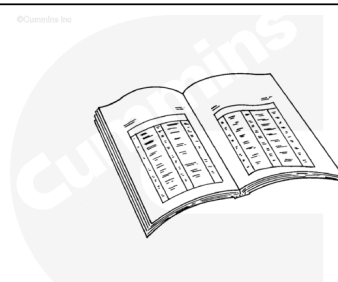
⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠



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State and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>)
- Clean the area around the lubricating oil cooler cover.
- Remove the coolant filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-007.html>)
- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/100/100-007-013.html>)
- Remove the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>)

Marine Applications

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Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

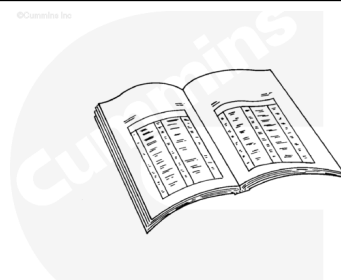
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⚠ CAUTION ⚠

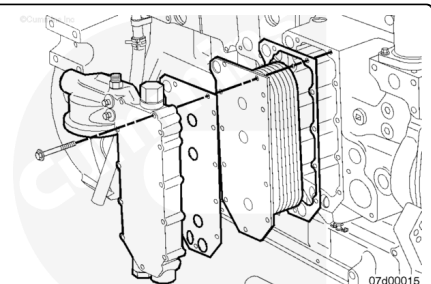
Use caution when draining oil or replacing filters that oil is not spilled or drained into the bilge area. The oil and oil filters must be discarded in accordance with local environmental regulations.

- Close the sea water supply and return valves. Refer to the original equipment manufacturer (OEM) service manual.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/45/45-008-018-tr.html>)
- Remove the heat exchanger/expansion tank assembly, if equipped. Refer to Procedure 008-053 in Section 8. (</qs3/pubsys2/xml/en/procedures/45/45-008-053-tr.html>)
- Clean the area around the lubricating oil cover.
- Remove the heat exchanger coolant return connection.
- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/45/45-007-013-tr.html>)
- Remove the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>)

Remove

Remove the lubricating oil cooler cover, gaskets and cooler element.

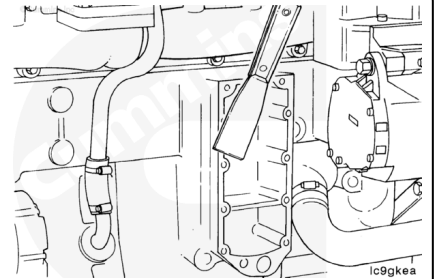
Discard the gaskets.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

Use a solvent that will not harm copper to clean the oil cooler elements.

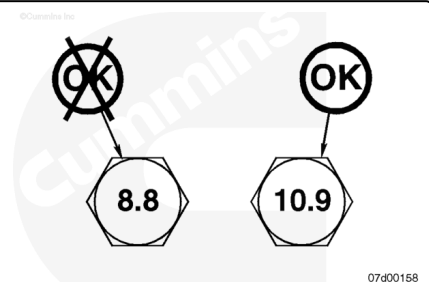
Use solvent to clean the oil cooler housing and cover.

Clean the sealing surfaces of the cylinder block and the oil cooler cover.

Note : Replace the elements if any debris is found or the engine has had debris-causing damage.

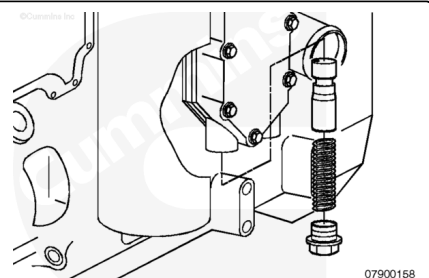
Inspect the capscrews. If they are Grade 10.9 or higher, they can be reused.

If the capscrews are lower than Grade 10.9, they **must** be replaced with Grade 10.9 or higher capscrews.



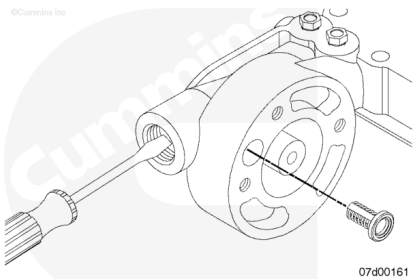
If any debris is suspected to have gone through the engine or if troubleshooting a lubricating oil pressure issue, remove and inspect the lubricating oil pressure regulator located in the lubricating oil cooler cover. Replace if necessary.

- For All Applications Except Marine: Refer to Procedure 007-029 in Section 7. (</qs3/pubsys2/xml/en/procedures/100/100-007-029.html>)
- For Marine Applications: Refer to Procedure 007-029 in Section 7. (</qs3/pubsys2/xml/en/procedures/45/45-007-029.html>)



If any debris is suspected to have gone through the engine, inspect the oil filter bypass valve located in the lubricating cooler cover. Make sure the valve is fully seated and opens and closes freely. Replace if necessary.

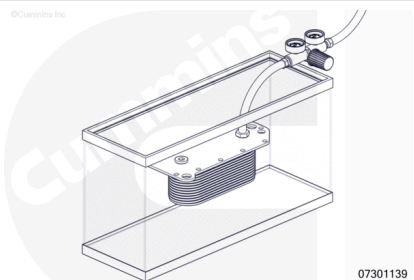
The bypass valve requires a 345 kPa [50 psi] pressure differential to open.



Leak Test

Use lubricating oil cooler pressure test kit, Part Number 3823876, to pressure-test the lubricating oil cooler element.

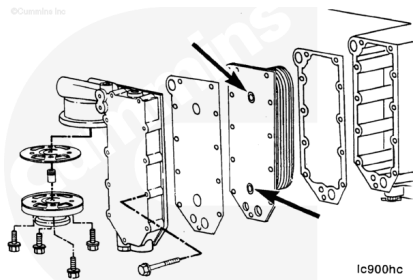
Air Pressure Test		
kpa		psi
449	MIN	65
518	MAX	75



Install

Note : Be sure to remove the shipping plugs from a new element.

Assemble the lubricating oil cooler gaskets, element, and cooler cover.

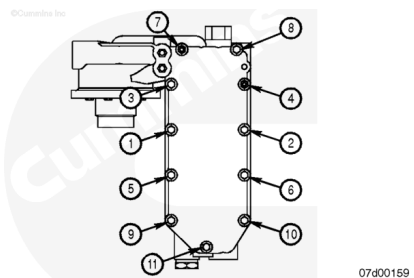


Tighten the capscrews in the sequence shown.

Torque Value:

- 1. 15 n•m [11 ft-lb]
- 2. 33 n•m [24 ft-lb]

Retorque capscrews 1 and 2 using the torque value listed below:

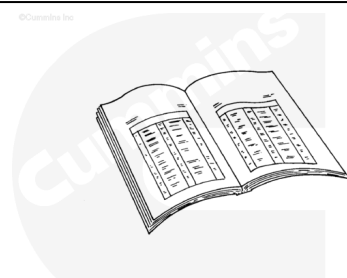


Torque Value: 33 n·m [24 ft·lb]

Finishing Steps

All Applications Except Marine

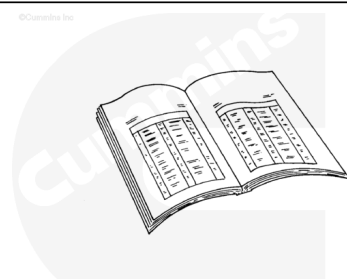
- Install the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>)
- Install the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/100/100-007-013.html>)
- Install the coolant filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-007.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>)
- Operate the engine and check for leaks.
- Stop the engine and check the coolant and lubricating oil level.



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Marine Applications

- Install the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>)
- Install the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/45/45-007-013-tr.html>)
- Install the heat exchanger/expansion tank, if equipped. Refer to Procedure 008-053 in Section 8. (</qs3/pubsys2/xml/en/procedures/45/45-008-053-tr.html>)
- Install the heat exchanger coolant return line.
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/45/45-008-018-tr.html>)
- Open the sea water supply line valves. Refer to the OEM service manual.
- Operate the engine and check for leaks.



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- Stop the engine and check the coolant and lubricating oil levels.

Last Modified: 02-Mar-2020
